

X-ray data and fit notes

Andrea Caputo,^{*a,b*}

Abstract. Sparse notes about X-ray data and fit.

Contents

1	Pyatomdb overview	1
2	Quick review of K-shell, L-shell, and M-shell Absorptions	3
3	Review of how elements abundances are determined	4

1 Pyatomdb overview

pyatomdb relies on the **Astrophysical Plasma Emission Code (APEC)** to model the emission from collisionally ionized diffuse gas. This emission is relevant for understanding X-ray spectra from various astrophysical sources, including galaxy clusters, supernova remnants, and the interstellar medium. The key assumption is that of Collisional Ionization Equilibrium (CIE):

- Assumes that the plasma is in a steady state where ionization and recombination rates are balanced.
- Characterized by a single temperature, T .

We have the following emission components.

Continuum Emission

- **Bremsstrahlung (Free-Free Emission)**: Emission of photons when electrons are decelerated by ions.

$$\frac{dW}{dVd\epsilon} \propto Z^2 n_e n_i T^{-1/2} e^{-\epsilon/kT} \quad (1.1)$$

where Z is the charge of the ion, n_e and n_i are the electron and ion densities, T is the temperature, and ϵ is the photon energy.

- **Recombination Continuum (Free-Bound Emission)**: Emission when a free electron is captured by an ion.

$$\frac{dW}{dVd\epsilon} \propto n_e n_i \alpha(\epsilon) e^{-\epsilon/kT} \quad (1.2)$$

where $\alpha(\epsilon)$ is the recombination coefficient.

- **Two-Photon Emission**: Emission of two photons during the decay of certain excited states (important for ions like He II).

Line Emission

- **Collisional Excitation**: Emission lines produced when electrons are excited to higher energy levels and then decay back.

$$j_{ul} = n_e n_i \frac{h\nu_{ul}}{4\pi} A_{ul} \quad (1.3)$$

where j_{ul} is the emissivity, ν_{ul} is the transition frequency, and A_{ul} is the Einstein A coefficient.

- The intensity of an emission line is given by:

$$I_{ul} = \int n_e n_i \frac{h\nu_{ul}}{4\pi} A_{ul} dV \quad (1.4)$$

These rates are essential for determining the ionization state of the plasma. They depend on the electron temperature and are usually obtained from atomic databases.

The energy levels of ions and the probabilities of transitions between these levels (Einstein A coefficients) are crucial for calculating line emission. Databases like AtomDB provide these values.

The relative abundances of elements in the plasma, typically compared to solar abundances, affect the line and continuum emission.

1. **Set Plasma Conditions:** Define the electron temperature T and the elemental abundances.
2. **Calculate Ionization Balance:** Determine the ionization fractions of each element in the plasma based on the temperature.
3. **Compute Emission Coefficients:** Use the ionization fractions, energy levels, and transition probabilities to calculate the emissivity for both continuum and line emissions.
4. **Integrate Over Volume:** If needed, integrate the emissivities over the volume of the plasma to get the total emitted power.
5. **Apply Redshift and Normalization:** Account for the redshift z of the source and apply a normalization factor related to the emission measure.

The final X-ray spectrum from the plasma is a combination of the continuum and line emissions, adjusted for redshift and normalized to the observed flux. The spectrum $S(E)$ in units of photons $\text{cm}^{-2} \text{s}^{-1} \text{keV}^{-1}$ is given by:

$$S(E) = \frac{(1+z)^{-1}}{4\pi D^2} \left(\int \epsilon_{\text{cont}}(E', T) dV + \sum \int \epsilon_{\text{line}}(E', T) dV \right) \quad (1.5)$$

where D is the distance to the source, $E' = E(1+z)$ is the redshifted energy, and ϵ_{cont} and ϵ_{line} are the continuum and line emissivities, respectively.

Summary

- **CIE Assumption:** Plasma is in a steady state with balanced ionization and recombination rates.
- **Continuum Emission:** Includes bremsstrahlung, recombination continuum, and two-photon emission.
- **Line Emission:** Includes collisional excitation lines from various ions.
- **Atomic Data:** Utilizes ionization rates, energy levels, transition probabilities, and elemental abundances from atomic databases.

- **Spectrum Calculation:** Combines emissivities from continuum and line emissions, adjusted for redshift and normalized to the observed flux.

The `pyatomdb` library leverages these principles to compute the emission spectrum of a hot, optically thin plasma, making it a powerful tool for interpreting X-ray observations in astrophysics.

2 Quick review of K-shell, L-shell, and M-shell Absorptions

In atomic physics, K-shell, L-shell, and M-shell absorptions refer to the absorption of photons by electrons in different energy levels or shells of an atom. These shells correspond to the principal quantum numbers $n = 1, 2$ and 3 respectively. Here's a brief overview of each:

K-shell Absorption

- **K-shell:** This is the innermost shell of an atom, corresponding to $n=1$.
- **Absorption Process:** When a photon with sufficient energy is absorbed, an electron in the K-shell can be ejected from the atom. This process requires a photon with energy greater than the binding energy of the K-shell electron.
- **Energy Range:** The binding energy for K-shell electrons varies greatly among different elements. For example, for hydrogen, it's 13.6 eV, but for heavier elements like iron, it's around 7.1 keV. An L-shell ($n = 2$) electron can then drop into the K-shell releasing 6.4 keV of energy

L-shell Absorption

- **L-shell:** This is the second shell of an atom, corresponding to $n=2$
- **Absorption Process:** Similar to the K-shell absorption, a photon with sufficient energy can eject an electron from the L-shell. The binding energy for L-shell electrons is lower than that for K-shell electrons.
- **Energy Range:** The binding energy for L-shell electrons is in the range of hundreds of eV to a few keV. For example, the L-shell binding energy for iron is around 0.72 keV.

M-shell Absorption

- **M-shell:** This is the third shell of an atom, corresponding to $n=3$.
- **Absorption Process:** Photons can also eject electrons from the M-shell if they have enough energy to overcome the binding energy of the electrons in this shell.
- **Energy Range:** The binding energies for M-shell electrons are typically lower, ranging from tens to hundreds of eV. For example, the M-shell binding energy for iron is around 0.1 keV.

Significance in Absorption Spectra

- **Absorption Edges:** Each shell has a characteristic absorption edge, which is the energy threshold above which photons can eject electrons from that shell. These edges appear as sudden increases in the absorption cross-section.
- **Element-Specific:** The exact energies of the absorption edges depend on the element. Heavier elements have higher binding energies and hence higher energy absorption edges.

Summary

- **K-shell Absorption:** Involves innermost electrons, high energy, and significant cross-section increases.
- **L-shell Absorption:** Involves second shell electrons, intermediate energies, noticeable absorption features.
- **M-shell Absorption:** Involves third shell electrons, lower energies, smaller absorption features compared to K and L shells.

Element	K (keV)	L (keV)	M (keV)
Aluminum	1.559	0.073	0.005
Argon	3.206	0.247	0.018
Calcium	4.038	0.344	0.023
Carbon	0.283	0.011	0.001
Chlorine	2.823	0.200	0.014
Chromium	5.989	0.574	0.054
Helium	0.0246	-	-
Hydrogen	0.0136	-	-
Iron	7.112	0.707	0.083
Magnesium	1.254	0.049	0.003
Neon	0.870	0.048	0.003
Nickel	8.333	0.853	0.112
Nitrogen	0.400	0.015	0.001
Oxygen	0.531	0.024	0.002
Silicon	1.840	0.099	0.007
Sodium	1.072	0.062	0.005
Sulfur	2.471	0.165	0.013

Table 1. K, L, M shell transition energies in keV for various elements

3 Review of how elements abundances are determined

Meteorites

Meteorites are valuable sources for determining the elemental abundances in the solar system. They are considered pristine samples of the early solar system, and their composition provides insights into the primordial material from which the solar system formed.

Elemental abundances are determined from meteorites through various analytical techniques, such as mass spectrometry and X-ray fluorescence. These methods allow precise measurements of the concentrations of different elements.

Challenges for Certain Elements

However, determining the abundances of certain elements from meteorites poses significant challenges. Elements like helium (He), carbon (C), oxygen (O), and hydrogen (H) are particularly difficult to measure due to several reasons:

- Helium is a highly volatile element, making it difficult to retain in solid materials. It can easily escape from meteorites, complicating its measurement.
- Carbon and oxygen are also challenging to measure accurately because they form various compounds and can be part of organic materials or trapped in minerals. These compounds can alter their apparent abundances.
- Hydrogen presents additional difficulties due to its volatility, chemical reactivity, and potential for contamination. Hydrogen is abundant in the Earth's atmosphere and hydrosphere, making contamination a significant issue. Techniques such as combustion analysis and isotope ratio mass spectrometry (IRMS) are employed to measure hydrogen, but these methods require careful control to avoid contamination.

Solar Spectroscopy

Given these challenges, alternative methods are used to complement meteorite data for certain elements, for example abundances of elements like hydrogen are often determined from solar spectroscopy. Observations of the Sun's photosphere provide relatively uncontaminated samples of the primordial solar nebula. Spectroscopic observations of the Sun's spectrum allow scientists to measure the abundance of hydrogen and other elements.